MCV4U: Calculus & Vectors — Review Package

Intersections of Lines & Planes / Systems of Equations

Complete step-by-step algebraic model solutions (No Matrices) attached at back.

Practice Questions

1. Determine whether the two 3D lines L_1 and L_2 intersect at a single point, are coincident, parallel and distinct, or skew:

$$L_1 : \vec{r} = (1, 3, -1) + t(2, 1, 4), \quad t \in \mathbb{R}$$

$$L_2 : \vec{r} = (4, -2, 2) + u(1, -1, 2), \quad u \in \mathbb{R}$$

If they intersect, calculate the exact coordinates of the point of intersection (POI).

2. Determine whether the two 3D lines L_1 and L_2 intersect at a single point, are coincident, parallel and distinct, or skew:

$$L_1 : \vec{r} = (2, 0, 1) + t(1, 3, -2), \quad t \in \mathbb{R}$$

$$L_2 : \vec{r} = (7, 1, -7) + u(-2, 1, 3), \quad u \in \mathbb{R}$$

If they intersect, calculate the exact coordinates of the point of intersection (POI).

3. Determine whether the two 3D lines L_1 and L_2 intersect at a single point, are coincident, parallel and distinct, or skew:

$$L_1 : \vec{r} = (0, 2, 5) + t(2, -4, 6), \quad t \in \mathbb{R}$$

$$L_2 : \vec{r} = (1, 1, 1) + u(-1, 2, -3), \quad u \in \mathbb{R}$$

4. Determine the exact point of intersection between the parametric line $x = 1 + 3t$, $y = 2 - t$, $z = -1 + 2t$, $t \in \mathbb{R}$ and the plane $2x + 3y - z - 11 = 0$.

5. Determine whether the line $\vec{r} = (3, 1, -2) + t(2, -1, 1)$, $t \in \mathbb{R}$ intersects the plane $x + 3y + z + 4 = 0$, lies in the plane, or is parallel and distinct from it.

6. Show algebraically that the line $\vec{r} = (1, 2, -1) + t(1, 1, 5)$, $t \in \mathbb{R}$ lies completely within the plane $2x + 3y - z - 9 = 0$.

7. Determine a parametric vector equation $\vec{r} = \vec{r}_0 + t\vec{d}$, $t \in \mathbb{R}$ for the line of intersection between the two non-parallel planes:

$$\pi_1 : x + 2y - z = 3 \quad \text{and} \quad \pi_2 : 2x - y + 3z = 1$$

8. Determine a parametric vector equation $\vec{r} = \vec{r}_0 + t\vec{d}$, $t \in \mathbb{R}$ for the line of intersection between the two non-parallel planes:

$$\pi_1 : 2x + y + z = 5 \quad \text{and} \quad \pi_2 : x - 2y + 2z = 4$$

9. Solve the following system of linear equations using algebraic elimination (do not use matrices). State the complete point of intersection $P(x, y, z)$:

$$\begin{aligned} x + y + z &= 5 \\ 2x - y + z &= 2 \\ x + 2y - z &= 3 \end{aligned}$$

10. Solve the following system of linear equations using algebraic elimination (do not use matrices). State the complete point of intersection $P(x, y, z)$:

$$2x + y - z = 4$$
$$x - 2y + 2z = -3$$
$$3x + 2y + z = 10$$

11. Solve the linear system algebraically (do not use matrices). Show that the system has infinitely many solutions and write the common line of intersection in parametric vector form $\vec{r} = \vec{r}_0 + t\vec{d}$, $t \in \mathbb{R}$:

$$x + y + z = 4$$
$$2x - y + 3z = 5$$
$$3x + 4z = 9$$

12. Algebraically determine whether the following system of three planes has a unique solution, infinitely many solutions, or no solution. If no solution, describe the geometric configuration:

$$x + y + z = 3$$
$$2x + 2y + 2z = 10$$
$$x - y + 2z = 4$$

13. Determine the exact point of intersection between the plane $2x - 3y + z + 1 = 0$ and the line passing through point $Q(1, 5, -2)$ that is perpendicular to the plane.

14. a) Determine the point of intersection between the plane $3x + y - 2z - 14 = 0$ and the line passing through the origin $O(0, 0, 0)$ that is perpendicular to the plane.

b) Determine the exact coordinates where the line $\vec{r} = (4, 6, -3) + t(2, 3, -1)$, $t \in \mathbb{R}$ intersects the xz-coordinate plane ($y = 0$).

15. Solve the system of three planes algebraically (do not use matrices):

$$\pi_1 : x + 2y - z = 4, \quad \pi_2 : 2x + 4y - 2z = 8, \quad \pi_3 : x - y + z = 1$$

Show that two planes are coincident and the third crosses, and write the line of intersection in parametric vector form $\vec{r} = \vec{r}_0 + t\vec{d}$, $t \in \mathbb{R}$.

16. Classify the geometric interaction of the three planes algebraically:

$$\pi_1 : x + y + z = 2, \quad \pi_2 : x + y + z = 5, \quad \pi_3 : 2x - y + z = 1$$

17. Calculate the acute angle θ (to the nearest degree) between the two intersecting planes:

$$\pi_1 : 2x - y + 2z - 5 = 0 \quad \text{and} \quad \pi_2 : x + 2y - 2z + 1 = 0$$

18. Determine the exact coordinates of the foot of the perpendicular N from the point $Q(2, 1, -2)$ to the 3D line $\vec{r} = (1, -1, 0) + t(2, 1, -1)$, $t \in \mathbb{R}$.

19. a) Consider the line $L : \frac{x-k}{2} = \frac{y+3}{1} = \frac{z-1}{3}$ and the plane $\pi : 2x - y - z - 5 = 0$. Determine the exact scalar value of $k \in \mathbb{R}$ such that line L lies completely within plane π.

b) Verify your result by testing whether direction vector $\vec{d}$ is perpendicular to normal $\vec{n}$ and checking whether point $P_0(k, -3, 1)$ lies in plane π.

20. a) Consider the two planes $\pi_1 : 3x - 6y + 9z + 4 = 0$ and $\pi_2 : 2x - 4y + 6z + k = 0$, where $k \in \mathbb{R}$. Determine the exact scalar value(s) of k for which:

i) Planes π_1 and π_2 do not intersect (parallel and distinct).

ii) Planes π_1 and π_2 intersect in a plane (coincident).

b) Explain why there is no value of k for which π_1 and π_2 intersect in a line.

Master Model Solutions (Algebraic Elimination Method — No Matrices)

1. Given: $L_1 : \vec{r} = (1, 3, -1) + t(2, 1, 4)$ and $L_2 : \vec{r} = (4, -2, 2) + u(1, -1, 2)$.

Step 1 (Check Direction Ratios): $\vec{m}_1 = (2, 1, 4)$ and $\vec{m}_2 = (1, -1, 2)$.
Ratio test: $\frac{2}{1} \neq \frac{1}{-1} \neq \frac{4}{2}$. The direction vectors are not scalar multiples ($\vec{m}_1 \nparallel \vec{m}_2$), so lines are not parallel or coincident.

Step 2 (Equate Components):

$$(1)\ 1 + 2t = 4 + u \implies 2t - u = 3$$

$$(2)\ 3 + t = -2 - u \implies t + u = -5$$

$$(3)\ -1 + 4t = 2 + 2u \implies 4t - 2u = 3$$

Step 3 (Solve 2×2 System): Add equations (1) and (2):

$$(2t - u) + (t + u) = 3 + (-5) \implies 3t = -2 \implies t = -\frac{2}{3}$$

Substitute $t = -\frac{2}{3}$ into (2):

$$-\frac{2}{3} + u = -5 \implies u = -5 + \frac{2}{3} = -\frac{13}{3}$$

Step 4 (Consistency Check): Substitute $t = -\frac{2}{3}$ and $u = -\frac{13}{3}$ into equation (3):

$$\text{LS} = 4\left(-\frac{2}{3}\right) - 2\left(-\frac{13}{3}\right) = -\frac{8}{3} + \frac{26}{3} = \frac{18}{3} = 6$$

$$\text{RS} = 3$$

Since LS $\neq$ RS ($6 \neq 3$), the system is inconsistent.

Conclusion: The lines do not intersect and are **skew lines**.

2. Given: $L_1 : \vec{r} = (2, 0, 1) + t(1, 3, -2)$ and $L_2 : \vec{r} = (7, 1, -7) + u(-2, 1, 3)$.

Step 1 (Check Direction Ratios): $\vec{m}_1 = (1, 3, -2)$ and $\vec{m}_2 = (-2, 1, 3)$. Non-parallel.

Step 2 (Equate Components):

$$(1)\ 2 + t = 7 - 2u \implies t + 2u = 5$$

$$(2)\ 3t = 1 + u \implies 3t - u = 1 \implies u = 3t - 1$$

$$(3)\ 1 - 2t = -7 + 3u \implies 2t + 3u = 8$$

Step 3 (Solve 2×2 System): Substitute $u = 3t - 1$ into equation (1):

$$t + 2(3t - 1) = 5 \implies t + 6t - 2 = 5 \implies 7t = 7 \implies t = 1$$

Then $u = 3(1) - 1 = 2$.

Step 4 (Consistency Check & POI): Substitute $t = 1$ and $u = 2$ into equation (3):

$$\text{LS} = 2(1) + 3(2) = 2 + 6 = 8, \quad \text{RS} = 8 \quad \text{(Consistent!)}$$

Find POI using $t = 1$ on L_1:

$$P = (2 + 1(1),\ 0 + 1(3),\ 1 + 1(-2)) = (3, 3, -1)$$

Conclusion: The lines intersect at a single point $\mathbf{P(3, 3, -1)}$.

3. Given: $L_1 : \vec{r} = (0, 2, 5) + t(2, -4, 6)$ and $L_2 : \vec{r} = (1, 1, 1) + u(-1, 2, -3)$.

Step 1 (Direction Vector Ratios): Notice $\vec{m}_1 = (2, -4, 6) = -2(-1, 2, -3) = -2\vec{m}_2$. The direction vectors are parallel ($\vec{m}_1 \parallel \vec{m}_2$).

Step 2 (Point Containment Test): Test if $P_0(0, 2, 5)$ from L_1 lies on L_2:

$$0 = 1 - u \implies u = 1$$

Substitute $u = 1$ into y-component of L_2: $y = 1 + 2(1) = 3 \neq 2$.

Conclusion: Lines have parallel directions but do not share points $\implies$ **Parallel and Distinct**.

4. Given: Line $x = 1 + 3t, y = 2 - t, z = -1 + 2t$ and Plane $2x + 3y - z - 11 = 0$.

Step 1 (Substitute Parametric Equations into Plane):

$$2(1 + 3t) + 3(2 - t) - (-1 + 2t) - 11 = 0$$

Step 2 (Solve for t):

$$(2 + 6t) + (6 - 3t) + (1 - 2t) - 11 = 0$$

$$(6t - 3t - 2t) + (2 + 6 + 1 - 11) = 0 \implies t - 2 = 0 \implies t = 2$$

Step 3 (Calculate POI Coordinates):

$$x = 1 + 3(2) = 7, \quad y = 2 - (2) = 0, \quad z = -1 + 2(2) = 3$$

Conclusion: **Point of Intersection** $P(7, 0, 3)$.

5. Given: Line $\vec{r} = (3, 1, -2) + t(2, -1, 1)$ and Plane $x + 3y + z + 4 = 0$.

Step 1 (Substitute into Plane): $x = 3 + 2t,\ y = 1 - t,\ z = -2 + t$.

$$(3 + 2t) + 3(1 - t) + (-2 + t) + 4 = 0$$

Step 2 (Simplify):

$$(2t - 3t + t) + (3 + 3 - 2 + 4) = 0 \implies 0t + 8 = 0 \implies 8 = 0 \quad \text{(Inconsistent!)}$$

Conclusion: No solution exists. The line is **parallel and distinct** from the plane.

6. Given: Line $\vec{r} = (1, 2, -1) + t(1, 1, 5)$ and Plane $2x + 3y - z - 9 = 0$.

Step 1 (Substitute into Plane): $x = 1 + t,\ y = 2 + t,\ z = -1 + 5t$.

$$2(1 + t) + 3(2 + t) - (-1 + 5t) - 9 = 0$$

Step 2 (Simplify):

$$(2 + 2t) + (6 + 3t) + (1 - 5t) - 9 = 0 \implies (2t + 3t - 5t) + (2 + 6 + 1 - 9) = 0$$

$$0t + 0 = 0 \implies 0 = 0 \quad \text{(True for all } t \in \mathbb{R})$$

Conclusion: The line **lies completely within the plane**.

7. Given: $\pi_1 : x + 2y - z = 3$ and $\pi_2 : 2x - y + 3z = 1$.

Step 1 (Direction Vector via Cross Product): $\vec{n}_1 = (1, 2, -1)$ and $\vec{n}_2 = (2, -1, 3)$.

$$\vec{d} = \vec{n}_1 \times \vec{n}_2 = (2(3) - (-1)(-1),\ (-1)(2) - 1(3),\ 1(-1) - 2(2)) = (5, -5, -5)$$

Scale direction vector by $\frac{1}{5}$: $\vec{d} = (1, -1, -1)$.

Step 2 (Find Point P_0 by Setting $z = 0$):

$$(1)\ x + 2y = 3, \quad (2)\ 2x - y = 1 \implies y = 2x - 1$$

Substitute into (1): $x + 2(2x - 1) = 3 \implies 5x - 2 = 3 \implies 5x = 5 \implies x = 1$. Then $y = 2(1) - 1 = 1$. Point $P_0(1, 1, 0)$.

Conclusion: $\tilde{\mathbf{r}} = (\mathbf{1}, \mathbf{1}, \mathbf{0}) + \mathbf{t}(\mathbf{1}, \mathbf{-1}, \mathbf{-1}), \quad \mathbf{t} \in \mathbb{R}$.

8. Given: $\pi_1 : 2x + y + z = 5$ and $\pi_2 : x - 2y + 2z = 4$.

Step 1 (Direction Vector): $\vec{n}_1 = (2, 1, 1)$ and $\vec{n}_2 = (1, -2, 2)$.

$$\vec{d} = \vec{n}_1 \times \vec{n}_2 = (1(2) - 1(-2),\ 1(1) - 2(2),\ 2(-2) - 1(1)) = (4, -3, -5)$$

Step 2 (Find Point P_0 by Setting $y = 0$):

$$(1)\ 2x + z = 5 \implies z = 5 - 2x$$

$$(2)\ x + 2z = 4 \implies x + 2(5 - 2x) = 4 \implies -3x + 10 = 4 \implies 3x = 6 \implies x = 2$$

Then $z = 5 - 2(2) = 1$. Point $P_0(2, 0, 1)$.

Conclusion: $\tilde{\mathbf{r}} = (\mathbf{2}, \mathbf{0}, \mathbf{1}) + \mathbf{t}(\mathbf{4}, \mathbf{-3}, \mathbf{-5}), \quad \mathbf{t} \in \mathbb{R}$.

9. Given System:

$$(1)\ x + y + z = 5 \quad (2)\ 2x - y + z = 2 \quad (3)\ x + 2y - z = 3$$

Step 1 (Eliminate Variable x): Multiply (1) by 2: $2x + 2y + 2z = 10$. Subtract (2):

$$(2x + 2y + 2z) - (2x - y + z) = 10 - 2 \implies (4)\ \mathbf{3y + z = 8}$$

Subtract (1) from (3):

$$(x + 2y - z) - (x + y + z) = 3 - 5 \implies (5)\ \mathbf{y - 2z = -2} \implies \mathbf{y = 2z - 2}$$

Step 2 (Solve 2×2 System for y and z): Substitute $y = 2z - 2$ into (4):

$$3(2z - 2) + z = 8 \implies 6z - 6 + z = 8 \implies 7z = 14 \implies z = 2$$

Then $y = 2(2) - 2 = 2$.

Step 3 (Back-Substitute to Find x): Substitute $y = 2$ and $z = 2$ into (1): $x+2+2 = 5 \implies x = 1$.

Conclusion: Unique point of intersection $\mathbf{P(1, 2, 2)}$.

10. Given System:

$$(1)\ 2x + y - z = 4 \quad (2)\ x - 2y + 2z = -3 \quad (3)\ 3x + 2y + z = 10$$

Step 1 (Eliminate Variable z): Add (1) and (3):

$$(2x + y - z) + (3x + 2y + z) = 4 + 10 \implies (4)\ \mathbf{5x + 3y = 14}$$

Multiply (1) by 2: $4x + 2y - 2z = 8$. Add to (2):

$$(4x + 2y - 2z) + (x - 2y + 2z) = 8 + (-3) \implies 5x = 5 \implies x = 1$$

Step 2 (Solve for y and z): Substitute $x = 1$ into (4): $5(1) + 3y = 14 \implies 3y = 9 \implies y = 3$. Substitute $x = 1, y = 3$ into (1): $2(1) + 3 - z = 4 \implies 5 - z = 4 \implies z = 1$.

Conclusion: Unique point of intersection $\mathbf{P(1, 3, 1)}$.

11. Given System:

$$(1)\ x + y + z = 4 \quad (2)\ 2x - y + 3z = 5 \quad (3)\ 3x + 4z = 9$$

Step 1 (Eliminate y using (1) and (2)): Add equation (1) and equation (2):

$$(x + y + z) + (2x - y + 3z) = 4 + 5 \implies 3x + 4z = 9$$

Notice this equation is **identical to Equation (3)**.

Step 2 (Parameterize the Infinite Solution Line): Equation (3) is redundant. Set parameter $z = t$. From $3x + 4z = 9 \implies 3x = 9 - 4t \implies x = 3 - \frac{4}{3}t$. From (1): $y = 4 - x - z = 4 - \left(3 - \frac{4}{3}t\right) - t = 1 + \frac{1}{3}t$.

Conclusion: Infinitely many solutions on line $\tilde{\mathbf{r}} = (\mathbf{3}, \mathbf{1}, \mathbf{0}) + \mathbf{s}(-\mathbf{4}, \mathbf{1}, \mathbf{3}), \quad \mathbf{s} \in \mathbb{R}$.

12. Given System:

$$(1)\ x + y + z = 3 \quad (2)\ 2x + 2y + 2z = 10 \quad (3)\ x - y + 2z = 4$$

Step 1 (Compare Equations (1) and (2)): Divide equation (2) by 2: $x + y + z = 5$. Equation (1) states $x + y + z = 3$, whereas equation (2) states $x + y + z = 5$. Subtracting gives $0 = 2$ (LS $\neq$ RS), which is a contradiction.

Conclusion: **No solution. Geometric configuration: Two parallel distinct planes cut by a third plane.**

13. Given: Plane $2x - 3y + z + 1 = 0$ and Point $Q(1, 5, -2)$.

Step 1 (Construct Normal Line Equation): Direction vector $\vec{d} = \vec{n} = (2, -3, 1)$. Line: $x = 1 + 2t, y = 5 - 3t, z = -2 + t$.

Step 2 (Substitute Line into Plane):

$$2(1 + 2t) - 3(5 - 3t) + (-2 + t) + 1 = 0$$

$$(2 + 4t) - (15 - 9t) + (-2 + t) + 1 = 0 \implies 14t - 14 = 0 \implies t = 1$$

Step 3 (Calculate POI): $x = 1 + 2(1) = 3,\ y = 5 - 3(1) = 2,\ z = -2 + 1 = -1$.

Conclusion: **Point of Intersection** $P(3, 2, -1)$.

14. *a) Plane $3x + y - 2z - 14 = 0$ and Line through $O(0, 0, 0)$:* Line: $x = 3t, y = t, z = -2t$. Substitute: $3(3t) + (t) - 2(-2t) - 14 = 0 \implies 9t + t + 4t - 14 = 0 \implies 14t = 14 \implies t = 1$. POI coordinates: $\mathbf{P(3, 1, -2)}$.

b) Intersect Line $\vec{r} = (4, 6, -3) + t(2, 3, -1)$ with xz-plane ($y = 0$): Set $y = 0 \implies 6 + 3t = 0 \implies 3t = -6 \implies t = -2$. $x = 4 + 2(-2) = 0,\ z = -3 + (-1)(-2) = -1$. Coordinates: $(\mathbf{0}, \mathbf{0}, -\mathbf{1})$.

15. Given System: $\pi_1 : x + 2y - z = 4, \quad \pi_2 : 2x + 4y - 2z = 8, \quad \pi_3 : x - y + z = 1$.

Step 1 (Compare π_1 and π_2): Multiply π_1 by 2: $2x + 4y - 2z = 8$, which matches π_2 identically $\implies$ π_1 and π_2 are coincident.

Step 2 (Solve System of π_1 and π_3):

$$(1)\ x+2y-z=4, \quad (3)\ x-y+z=1$$

Add (1) and (3): $2x+y=5 \implies y=5-2x$. Let $x=t \implies y=5-2t$. From (3): $z=1-x+y=1-t+(5-2t)=6-3t$.

Conclusion: π_1 **and** π_2 **are coincident; common line** $\vec{\mathbf{r}}=(\mathbf{0},\mathbf{5},\mathbf{6})+\mathbf{t}(\mathbf{1},\mathbf{-2},\mathbf{-3}), \quad \mathbf{t}\in\mathbb{R}$.

16. Given System: $\pi_1: x+y+z=2, \quad \pi_2: x+y+z=5, \quad \pi_3: 2x-y+z=1$.

Step 1 (Examine Normals): $\vec{n}_1=(1,1,1)=\vec{n}_2$. Planes π_1 and π_2 have identical normals but $2\neq 5 \implies$ parallel and distinct. $\vec{n}_3=(2,-1,1)$ is non-parallel to $\vec{n}_1$.

Conclusion: **No solution. Configuration: Two parallel distinct planes intersected by a third plane.**

17. Given Planes: $\pi_1: 2x-y+2z-5=0$ and $\pi_2: x+2y-2z+1=0$.

Step 1 (Extract Normals & Dot Product): $\vec{n}_1=(2,-1,2)$ and $\vec{n}_2=(1,2,-2)$.

$$\vec{n}_1\cdot\vec{n}_2=(2)(1)+(-1)(2)+(2)(-2)=2-2-4=-4 \implies |\vec{n}_1\cdot\vec{n}_2|=4$$

Step 2 (Magnitudes):

$$|\vec{n}_1|=\sqrt{4+1+4}=3, \quad |\vec{n}_2|=\sqrt{1+4+4}=3$$

Step 3 (Calculate Angle θ):

$$\cos\theta=\frac{|\vec{n}_1\cdot\vec{n}_2|}{|\vec{n}_1||\vec{n}_2|}=\frac{4}{3\times 3}=\frac{4}{9}$$

$$\theta=\cos^{-1}\left(\frac{4}{9}\right)\approx 63.61^\circ \implies \mathbf{64^\circ}$$

18. Given Point $Q(2,1,-2)$ and Line $\vec{r}=(1,-1,0)+t(2,1,-1)$.

Step 1 (General Point $N(t)$ on Line): $N(t)=(1+2t,\ -1+t,\ -t)$.

Step 2 (Vector $\vec{QN}$):

$$\vec{QN}=N-Q=((1+2t)-2,\ (-1+t)-1,\ (-t)-(-2))=(2t-1,\ t-2,\ 2-t)$$

Step 3 (Orthogonality Condition $\vec{QN}\cdot\vec{d}=0$): Direction vector $\vec{d}=(2,1,-1)$.

$$(2t-1)(2)+(t-2)(1)+(2-t)(-1)=0$$

$$(4t-2)+(t-2)+(t-2)=0 \implies 6t-6=0 \implies t=1$$

Step 4 (Coordinates of N): $x=1+2(1)=3,\ y=-1+1=0,\ z=-(1)=-1$.

Conclusion: **Foot of Perpendicular** $N(3,0,-1)$.

19. *a) Line $L: \frac{x-k}{2}=\frac{y+3}{1}=\frac{z-1}{3}$ in Plane $2x-y-z-5=0$:* Line direction $\vec{d}=(2,1,3)$, normal $\vec{n}=(2,-1,-1)$. $\vec{d}\cdot\vec{n}=2(2)+1(-1)+3(-1)=4-1-3=0$ (Parallel). Initial point $P_0(k,-3,1)$ must satisfy plane:

$$2(k)-(-3)-(1)-5=0 \implies 2k+3-1-5=0 \implies 2k-3=0 \implies \mathbf{k=\frac{3}{2}}$$

b) Verification: $\vec{d} \cdot \vec{n} = 0$ confirms line direction is parallel to plane. Plugging $k = \frac{3}{2}$ into plane: $2(\frac{3}{2}) - (-3) - 1 - 5 = 3 + 3 - 1 - 5 = 0$. Confirmed.

20. *a) Planes* $\pi_1 : 3x - 6y + 9z + 4 = 0$ *and* $\pi_2 : 2x - 4y + 6z + k = 0$*:* $\vec{n}_1 = (3, -6, 9) = \frac{3}{2}(2, -4, 6) = \frac{3}{2}\vec{n}_2 \implies$ Normals are parallel. Multiply π_2 by $\frac{3}{2}$: $3x - 6y + 9z + \frac{3}{2}k = 0$. For coincident planes, constants match: $\frac{3}{2}k = 4 \implies k = \frac{8}{3}$.

i) Parallel and Distinct: $\mathbf{k \in \mathbb{R},\ k \neq \frac{8}{3}}$.

ii) Coincident: $\mathbf{k = \frac{8}{3}}$.

b) Explanation: Because normal vectors $\vec{n}_1 \parallel \vec{n}_2$ for all $k \in \mathbb{R}$, the planes are always parallel. Parallel planes can never intersect along a line.